

WEB APPLICATION VULNERABILITY REPORT

Cyber Security Internship – Task 1

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Target: <http://testphp.vulnweb.com>

i. Executive Summary

This report presents the results of a vulnerability assessment conducted on the test web application 'testphp.vulnweb.com'. The objective of the assessment was to identify potential security weaknesses in the application using automated and manual security testing techniques. The assessment utilized tools such as Nmap for reconnaissance and OWASP ZAP for web vulnerability scanning.

ii. Target Information

Target Website: <http://testphp.vulnweb.com>

Application Type: Test Web Application

Purpose: Security Testing Environment

Site screenshot below:



iii. Tools Used

- Nmap (Zenmap) – Network reconnaissance and service discovery
- OWASP ZAP – Automated vulnerability scanning
- Browser Developer Tools – Manual inspection of network requests and headers

iv. Methodology

Step 1: Reconnaissance using Nmap to identify open ports and services.

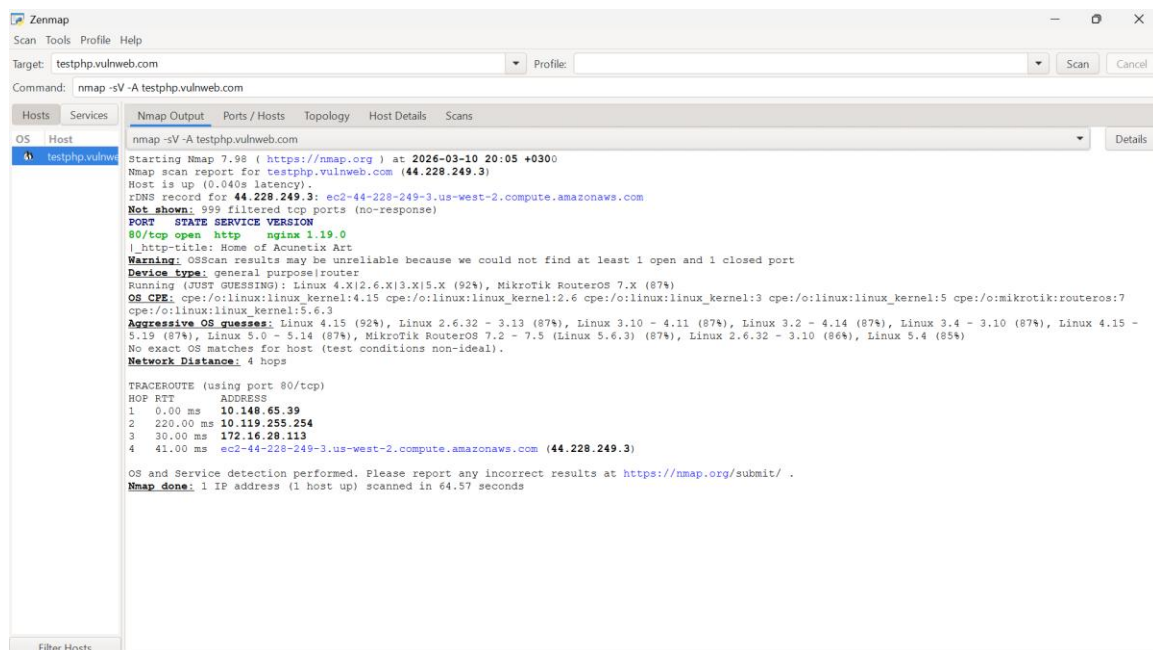
Step 2: Automated vulnerability scanning using OWASP ZAP.

Step 3: Manual verification using browser developer tools.

Step 4: Classification of vulnerabilities based on severity levels.

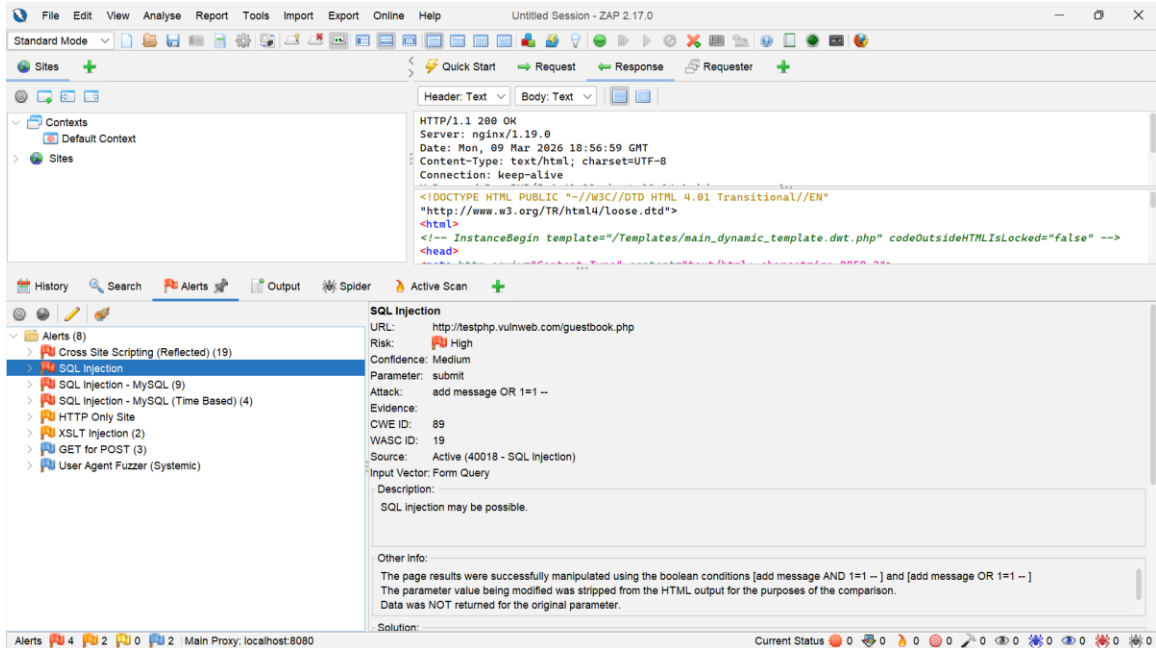
v. Nmap Scan Results

Insert Nmap scan screenshot below:



vi. OWASP ZAP Vulnerability Findings

Insert OWASP ZAP scan results screenshot below:

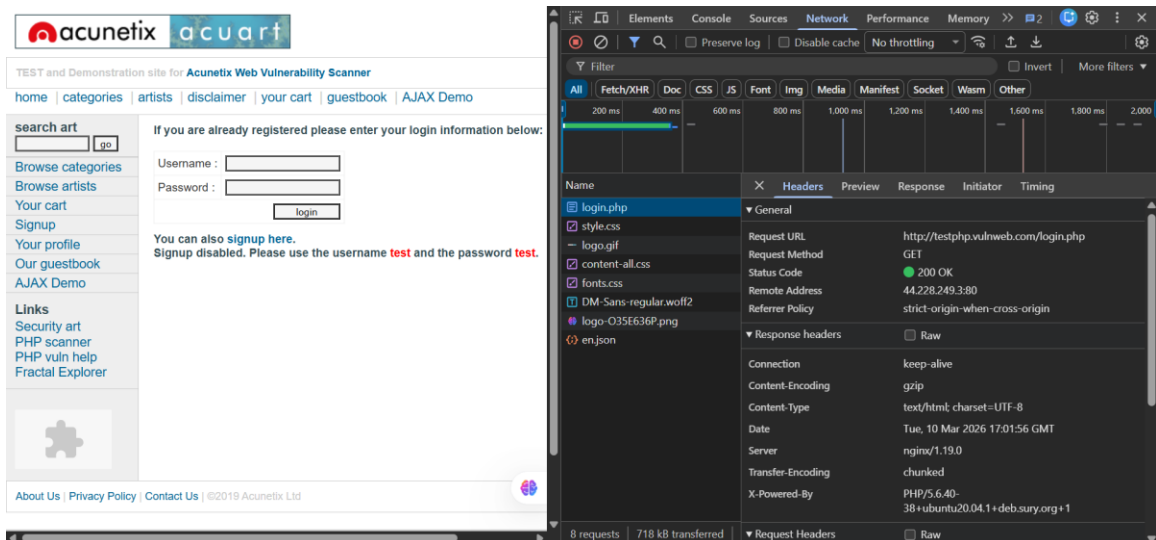


Example vulnerabilities identified:

- SQL Injection
- Cross Site Scripting (XSS)
- Missing Security Headers

vii. Manual Inspection using Browser Developer Tools

Insert browser developer tools screenshot below:



viii. Risk Classification

High Risk: SQL Injection vulnerabilities

Medium Risk: Cross Site Scripting (XSS)

Low Risk: Missing security headers or information disclosure

ix. Remediation Recommendations

- Implement input validation and parameterized queries to prevent SQL Injection.
- Use proper output encoding to mitigate Cross Site Scripting vulnerabilities.
- Implement secure HTTP headers such as Content-Security-Policy and X-Frame-Options.
- Regularly perform vulnerability scans and security audits.

x. Conclusion

The vulnerability assessment demonstrated how common security testing tools can be used to identify weaknesses in web applications. The findings highlight the importance of secure coding practices and regular security testing.